AI-Driven Crop Disease Detection Using Image Processing and Computer Vision Techniques



By

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Project Card

Start Date and End Date of the project

1 December 2025 – 30 September 2026

Total Sanctioned Amount

₹ 2,40,000

Total Amount Disbursed from TIH

₹ 2,40,000

Awards/Recognition

Attended AI4Agri Conference 2026; collaborative dataset procurement from the PoCRA team

Action Items from the previous review

April 2026 review was the 1st review (no prior action items). Planned: pilot deployment for field testing and integration for end-to-end prediction.

IP/ Patents details

IP of models and dataset belongs to TIH IIT Bombay. No patents filed (Nil). Paper submitted to IEEE PerCom 2027.

Commercialization Plan

Not applicable at this stage. Pilot deployment and application integration are planned next steps.

Outline

- **Background, Problem Statement**
- **Objectives**
- **Expected Benefits of the Project**
- **Major Milestones**
- **Shortfalls and reasons in Achievements of Milestones**
- **Technical/Scientific Achievements for the Review Period**
- **List of Publications, Patents etc.**
- **Future Work Planned**
- **Conclusions**

Background & Problem Statement

- Crop disease is a major controllable yield loss for smallholder farmers in Maharashtra; manual field inspection does not scale.
- This project builds image-based disease classifiers for **Sugarcane, Pigeon Pea, Grapes and Mustard** under the CHANAKYA Fellowship, supporting the **MahaVISTAAR** AI-based advisory platform.
- **Problem:** models trained on controlled PoCRA imagery are usually judged by internal accuracy, yet may fail on real field photographs (NPSS) – a domain shift.
- **Approach:** supervised transfer learning (ResNet, DenseNet, EfficientNet, ConvNeXt, CSPDarkNet, Swin, DINOv2 and others) fine-tuned on PoCRA and tested on NPSS.

60+1

experiment configurations (23 Sugarcane, 29 Pigeon Pea, 8 Grapes) + 1 Mustard model

24–39%

NPSS accuracy of Sugarcane baselines that scored 85–97% internally

4 crops

PoCRA-trained, NPSS field-evaluated (Mustard pending)

Objectives

#	Objective	Status
1	Develop image-based disease-classification models for Sugarcane, Pigeon Pea, Grapes and Mustard	Achieved
2	Evaluate PoCRA→NPSS cross-domain generalisation	Partially achieved – Mustard pending; Grapes indicative (42 images)
3	Identify interventions that narrow the internal-to-field accuracy gap and test whether internal accuracy predicts field accuracy	Achieved
4	Document methodology, results and reproducibility artifacts	Partially achieved – recorded at directory level (VDI-22)
5	Audit deployed models on NPSS data for additional crops and build NPSS data-extraction tooling (June 2026)	Achieved

Expected Benefits of the Project

Farmers

- Scalable image-based diagnosis in place of manual inspection
- Delivered through the MahaVISTAAR advisory platform
- Field-realistic accuracy expectations (66–79% best on NPSS)

TIH IIT Bombay / PoCRA

- Trained models for four crops
- A repeatable PoCRA→NPSS evaluation process
- Audit of deployed models on NPSS for additional crops

Researchers

- Documented 60+ experiments with metrics
- Evidence that internal accuracy is an unreliable guide to field accuracy
- Artifact index and a paper submitted to IEEE PerCom 2027

Beneficiaries: smallholder farmers in Maharashtra; TIH IIT Bombay; PoCRA; future researchers and fellows.

Major Milestones

#	Milestone	Target	Actual	Status
1	Project set-up and dataset understanding	Dec 2025	Dec 2025	Completed
2	Data-processing pipeline; baseline on raw dataset	Jan 2026	Jan 2026	Completed
3	PoCRA dataset integration; crop-specific (Pigeon Pea, Mustard) training	Feb 2026	Feb 2026	Completed
4	Application testing	Feb 2026	Feb 2026	Completed
5	GitHub collaboration and Docker fundamentals	Mar 2026	Mar 2026	Completed
6	Sugarcane study (23 configs); June deployed-model NPSS audit	–	Jun 2026	Completed
7	Pigeon Pea (29) and Grapes (8) studies; Mustard model; Handover Report; paper	–	Jul–Sep 2026	Completed

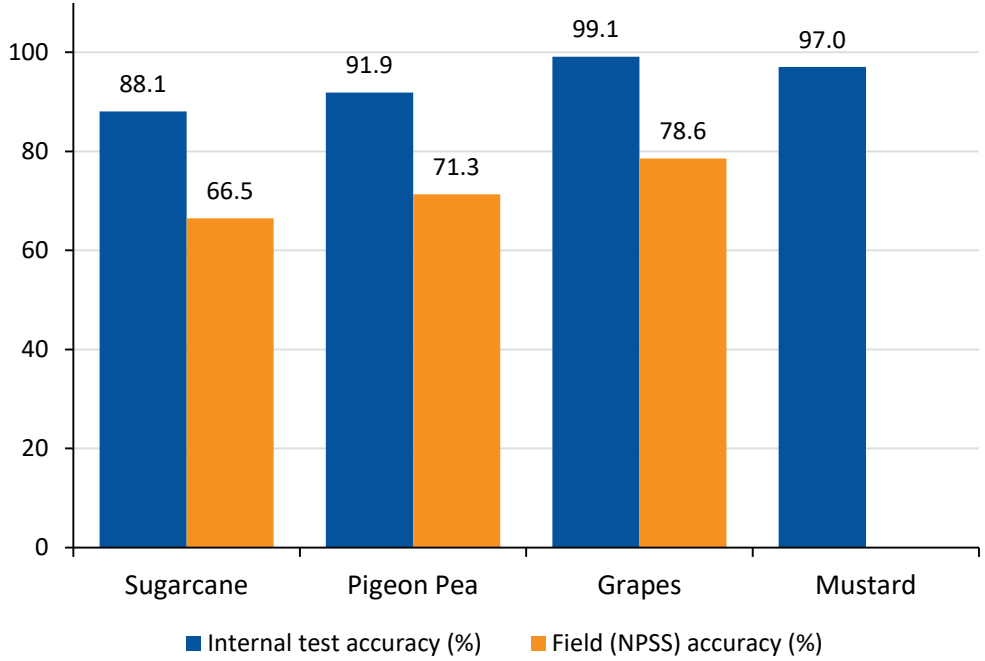
Milestones 1–5: April 2026 review presentation. Funds: ₹2,40,000 sanctioned and disbursed.

Shortfalls and Reasons

All approved milestones were completed on their target dates. Open items are limits of scope:

Open item	Reason / detail	Next step
Mustard: no NPSS evaluation	Only the internal PoCRA-curated test set (97.02%) was evaluated	Future work
Grapes: NPSS set of only 42 images	Small validated sample – results are indicative, not robust	Future work
Pigeon Pea Exp 10, 11, 16 not on NPSS	ShuffleNet and LiteMDC configurations were not evaluated on NPSS	Future work
Reproducibility artifacts at directory level	Splits, checkpoints and code on VDI-22; file-level split, seed and commit not recorded	Recording standard
Single run per configuration	Computational constraints; no confidence intervals	Repeated trials

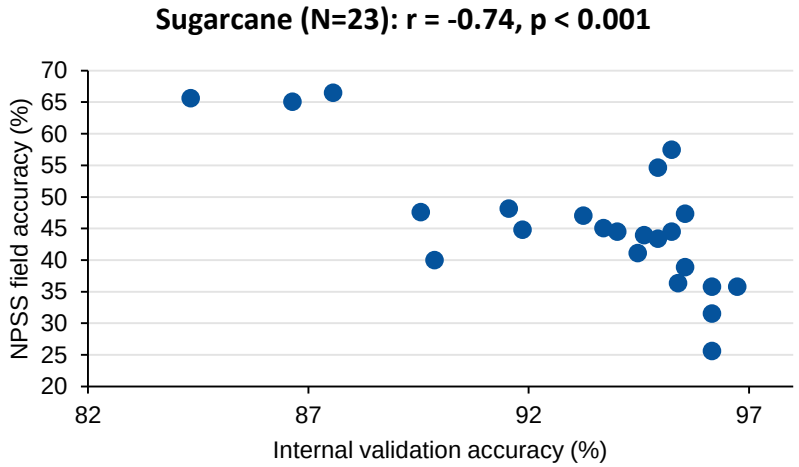
Achievements – Headline Results

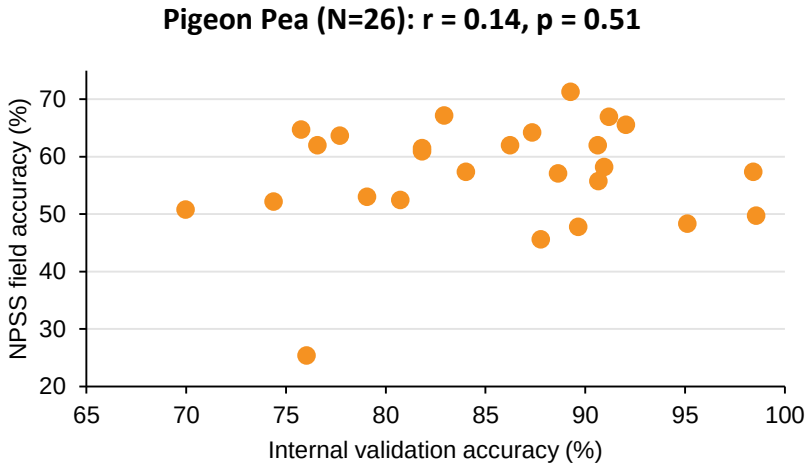


Best field-accuracy configuration per crop. Grapes field result is on 42 NPSS images; Mustard has no NPSS evaluation.

- **Sugarcane:** CSPDarkNet53 + dropout/BN-freeze + class weights – **66.48%** NPSS (baselines 31.55–38.87%)
- **Pigeon Pea:** ConvNeXt-Base + augmentation + SAM – **71.31%** NPSS
- **Grapes:** DINOv2 + augmentation + SAM – **78.57%** NPSS (99.21% validation)
- **Mustard:** ResNet50 – **97.02%** internal test accuracy, macro F1 ≈ 0.967
- Best Sugarcane foundation model: DINOv2 + class weights, 57.46%

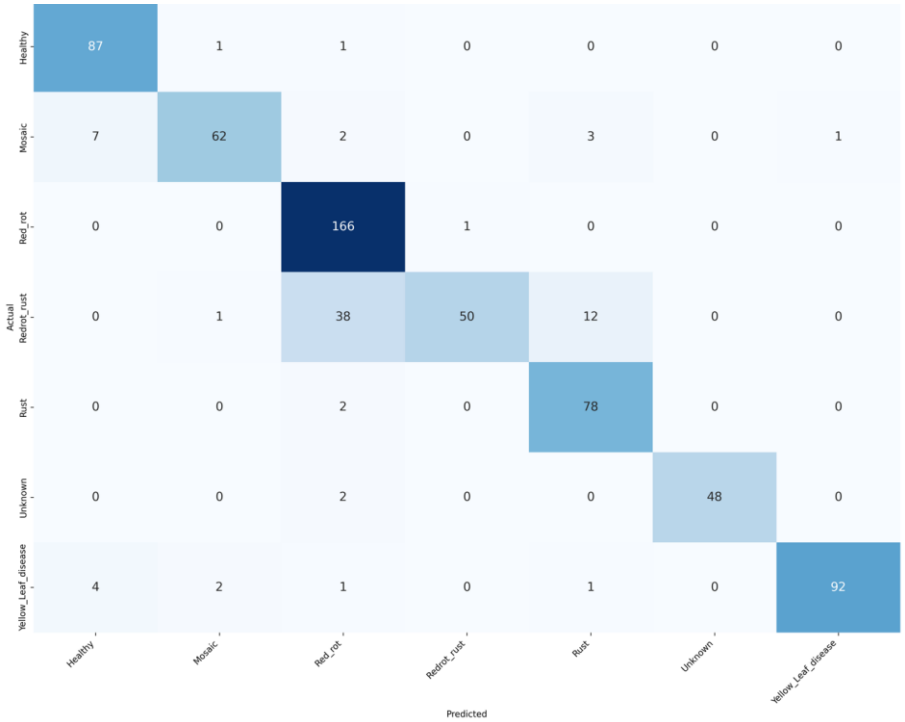
Achievements – Internal vs Field





- **Internal accuracy is not a reliable proxy for field accuracy:** negatively correlated for Sugarcane (Spearman ρ = -0.73), uncorrelated for Pigeon Pea (ρ = 0.05).
- **Sugarcane:** class-weighted loss added +21 to +30 NPSS points for 7–10 points of internal accuracy.
- **Pigeon Pea:** SAM on ConvNeXt-Base cost under 2 internal points for over 4 NPSS points.

Sugarcane – Best Field Model (Exp 40)



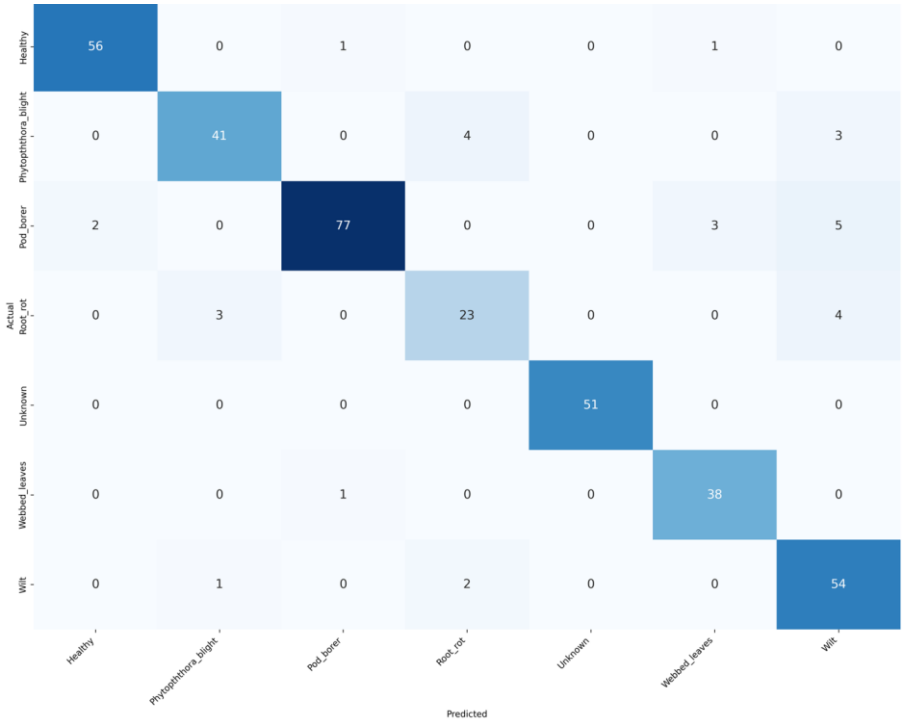
Confusion matrix (test set) – rows: actual, columns: predicted

Exp 40: CSPDarkNet53 + Dropout/BN Freeze + Class Weights

Class	Precision	Recall	F1	Support
Healthy	0.8878	0.9775	0.9305	89
Mosaic	0.9394	0.8267	0.8794	75
Red_rot	0.7830	0.9940	0.8760	167
Redrot_rust	0.9804	0.4950	0.6579	101
Rust	0.8298	0.9750	0.8966	80
Unknown	1.0000	0.9600	0.9796	50
Yellow_Leaf_disease	0.9892	0.9200	0.9534	100
Accuracy			0.8807	662
Macro avg	0.9157	0.8783	0.8819	662
Weighted avg	0.8981	0.8807	0.8724	662

- Best field model: **66.48%** NPSS accuracy; **88.07%** internal test accuracy (662 test images, macro F1 0.8819).
- Weakest class: **Redrot_rust** (recall 0.4950, F1 0.6579); Red_rot recall 0.9940 but precision 0.7830.

Pigeon Pea – Best Field Model (Exp 12)



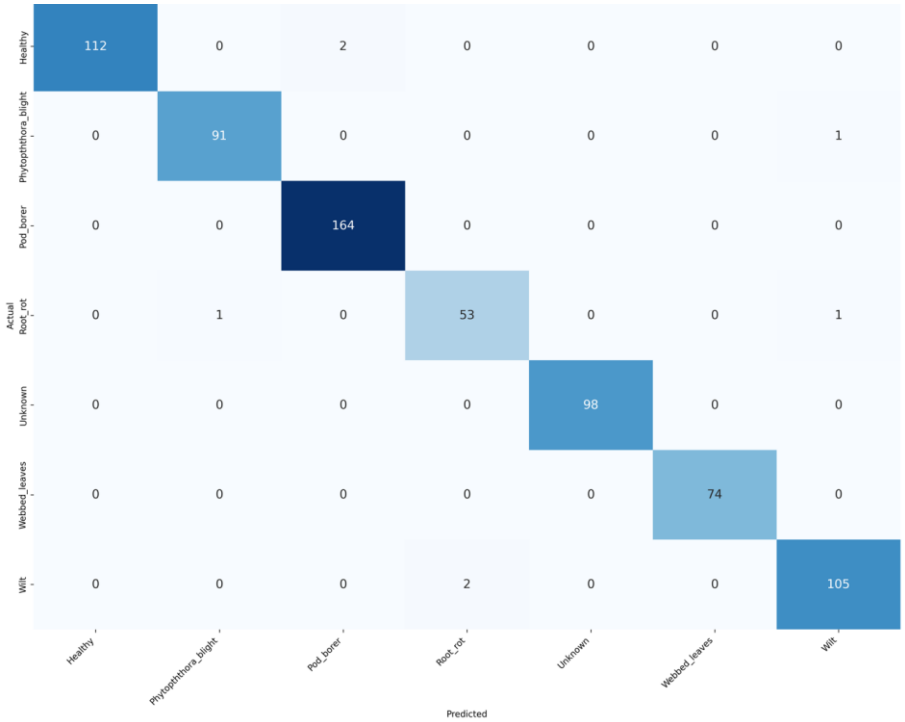
Confusion matrix (test set) – rows: actual, columns: predicted

Exp 12: ConvNeXt-Base + Augmentation + SAM

Class	Precision	Recall	F1	Support
Healthy	0.9655	0.9655	0.9655	58
Phytopthora_blight	0.9111	0.8542	0.8817	48
Pod_borer	0.9747	0.8851	0.9277	87
Root_rot	0.7931	0.7667	0.7797	30
Unknown	1.0000	1.0000	1.0000	51
Webbed_leaves	0.9048	0.9744	0.9383	39
Wilt	0.8182	0.9474	0.8780	57
Accuracy			0.9189	370
Macro avg	0.9096	0.9133	0.9101	370
Weighted avg	0.9223	0.9189	0.9191	370

- Best Pigeon Pea field model: **71.31%** NPSS accuracy; **91.89%** internal test accuracy (macro F1 0.9101).
- Weakest class: **Root_rot** (23 of 30 correct, F1 0.7797); Unknown is classified perfectly (51 of 51).

Pigeon Pea – Highest Internal Accuracy (Exp 22)



Confusion matrix (test set) – rows: actual, columns: predicted

Exp 22: Swin Transformer + Augmentation + CLAHE

Class	Precision	Recall	F1	Support
Healthy	1.0000	0.9825	0.9912	114
Phytopthora_blight	0.9891	0.9891	0.9891	92
Pod_borer	0.9880	1.0000	0.9939	164
Root_rot	0.9636	0.9636	0.9636	55
Unknown	1.0000	1.0000	1.0000	98
Webbed_leaves	1.0000	1.0000	1.0000	74
Wilt	0.9813	0.9813	0.9813	107
Accuracy			0.9901	704
Macro avg	0.9889	0.9881	0.9885	704
Weighted avg	0.9901	0.9901	0.9901	704

- Highest internal test accuracy (**99.01%**, macro F1 0.9885) but only **49.73%** on NPSS field images.
- Illustrates the internal-to-field gap: near-perfect internal scores did not carry over to field data.

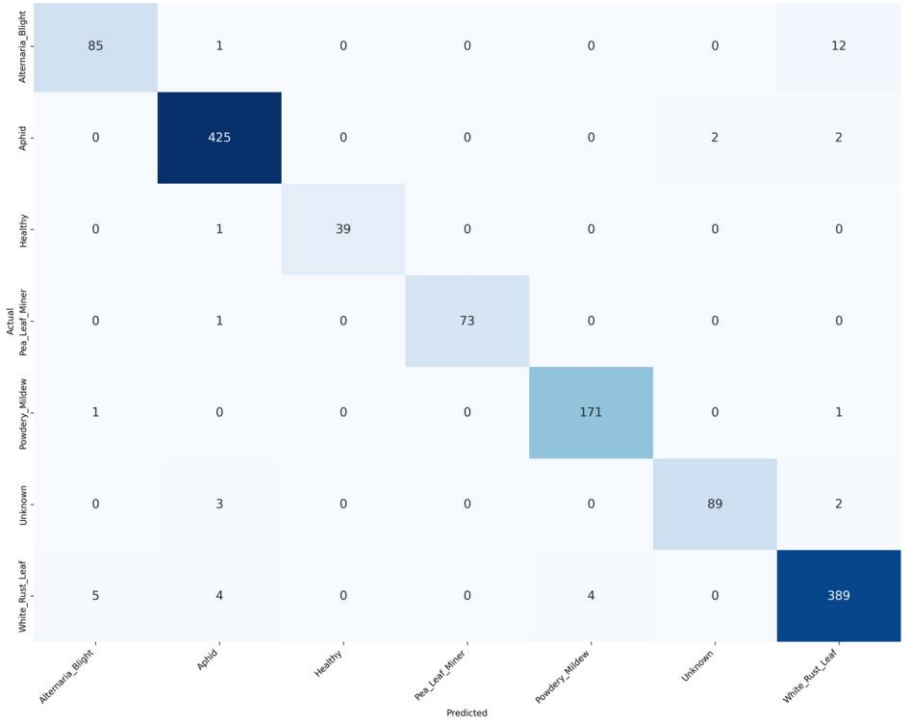
Pigeon Pea – ResNet50 Baseline vs Two-Phase

Aspect	Baseline (resnet50.pth)	Final (resnet50_4.pth)
Training strategy	Single phase; backbone frozen, FC only	Two phase: frozen, then unfrozen backbone
Learning rate	Single LR throughout	Phase 2 uses a lower LR
Test accuracy	86.52%	88.41% (+1.89)
Macro precision / recall / F1	0.8375 / 0.8500 / 0.8413	0.8643 / 0.8575 / 0.8595

Class	Baseline P / R / F1	Final P / R / F1
Healthy	0.818 / 0.931 / 0.871	0.900 / 0.931 / 0.915
Phytopththora_blight	0.886 / 0.813 / 0.848	0.870 / 0.833 / 0.851
Pod_borer	0.926 / 0.862 / 0.893	0.868 / 0.908 / 0.888
Root_rot	0.513 / 0.667 / 0.580	0.656 / 0.700 / 0.677
Unknown	1.000 / 1.000 / 1.000	1.000 / 1.000 / 1.000
Webbed_leaves	0.923 / 0.923 / 0.923	0.875 / 0.718 / 0.789
Wilt	0.796 / 0.754 / 0.775	0.881 / 0.912 / 0.897

Source: April 2026 review (early CROP06 iteration; the final model corresponds to Exp 2 in the consolidated study).

Mustard – Model Results (ResNet50)



Confusion matrix (test set) – rows: actual, columns: predicted

ResNet50 – PoCRA-curated Mustard test set

Class	Precision	Recall	F1	Support
Alternaria_Blight	0.9341	0.8673	0.8995	98
Aphid	0.9770	0.9907	0.9838	429
Healthy	1.0000	0.9750	0.9873	40
Pea_Leaf_Miner	1.0000	0.9865	0.9932	74
Powdery_Mildew	0.9771	0.9884	0.9828	173
Unknown	0.9780	0.9468	0.9622	94
White_Rust_Leaf	0.9581	0.9677	0.9629	402
Accuracy			0.9702	1310
Macro avg	0.9749	0.9603	0.9674	1310
Weighted avg	0.9701	0.9702	0.9700	1310

- Overall test accuracy **97.02%** (1,310 images); NPSS field evaluation for Mustard is pending.
- Weakest class: **Alternaria_Blight** (recall 0.8673) – 12 of 98 images were predicted as White_Rust_Leaf.

Achievements – June 2026 NPSS Audit

TIH Foundation for IoT & IoE

Deployed-model audit on NPSS

- NPSS data-extraction script pushed to GitHub (PoCRA_Test_Scripts) with a pull request
- NPSS datasets extracted for Cotton, Soybean, Chickpea, Cucumber, Onion, Grapes, Sugarcane, Maize and Papaya
- Crop-wise accuracy, per-class and misclassification reports prepared
- Also evaluated: Sorghum, Tomato, Mango, and threshold testing on Cotton

26.11%

Cotton – deployed model, NPSS overall accuracy

36.82%

Soybean – deployed ResNet-50, NPSS overall accuracy

Sugarcane improvement (15–19 June)

- ResNet50/34 and MobileNetV2 trained and evaluated on NPSS at thresholds 0.00% and 70.00%
- Class weights, dropout + BN-freeze, Mixup + SAM, then EfficientNet-B7, DenseNet201, ConvNeXt-Base
- These runs fed the 23-configuration Sugarcane study

Publications and Patents

Title	Venue	Status	Authors
From Lab to Field: Diagnosing the Internal-to-Field Accuracy Gap in Sugarcane and Pigeon Pea Disease Classification	IEEE PerCom 2027	Submitted	Yadnika Pashte, Yashodhan Rajapkar, Dr. Sushopti Gawade, Susmita Banerjee, Prajwal Rameshrao Asole

Paper highlights

- 52 experiments (23 Sugarcane, 29 Pigeon Pea)
- Best field accuracy 66.48% and 71.31%
- Review in progress; conference: 8–12 March 2027, Goa, India

Patents

- Nil – no patent was filed during the project (1 Dec 2025 – 30 Sep 2026)
- Handover Report v1.0 prepared 15 Sep 2026

Future Work Planned

TIH Foundation for IoT & IoE

1 Complete NPSS coverage

NPSS evaluation for Mustard; larger validated NPSS sample for Grapes

2 Domain adaptation

Correlation alignment, adversarial domain-invariant learning, Deep CORAL

3 Class weights + backbones

Pair class-weighted loss with ConvNeXt-Base, Swin, DINOv2 for Sugarcane

4 Pigeon Pea improvements

Test-time augmentation, few-shot fine-tuning, pseudo-label self-training

5 Robust evaluation

Repeated trials with confidence intervals; record split, seed, checkpoint, commit for every run

6 Deployment

Integrate models into the production application; real-world testing and monitoring

Conclusions

- Internal accuracy is **not** a trustworthy predictor of field accuracy: negatively correlated (Sugarcane) or uncorrelated (Pigeon Pea).
- Domain-gap levers differ by crop: **class-weighted loss** for Sugarcane, **SAM** for Pigeon Pea; ConvNeXt-Base and DINOv2 generalised most consistently.
- Grapes (42-image NPSS set) reached 78.57%; Mustard reached 97.02% internally with NPSS evaluation outstanding.
- Best field accuracies (66–79%) remain below internal accuracy – cross-domain evaluation should be standard practice before deployment.

66.48%

Sugarcane – best NPSS accuracy

71.31%

Pigeon Pea – best NPSS accuracy

78.57%

Grapes – best NPSS accuracy (42 images)